

Survival Analysis

Homework 3

Samvel Stepanyan

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Libraries Used

Important Note: `survminer` package requires `survMisc`, `KMsurv`, `km.ci`. `SurvMisc` is required for `survminer`, though you need to download the latest version manually from <https://cran.r-project.org/src/contrib/Archive/survMisc/>, download the 2022 version.

```
install.packages(c("KMsurv", "km.ci"), repos = "https://cloud.r-project.org")
```

#Specify the path if you downloaded `survMisc` anywhere else than Downloads folder

```
install.packages(  
  "~/Downloads/survMisc_0.5.6.tar.gz",  
  repos = NULL,  
  type = "source"  
)
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2     3.5.2      v tibble     3.2.1  
## v lubridate  1.9.4      v tidyr      1.3.1  
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(survival)
```

```
library(survminer)
```

```
## Loading required package: ggpubr
```

```
##
```

```
## Attaching package: 'survminer'
```

```
##
```

```
## The following object is masked from 'package:survival':
```

```
##
```

```
##      myeloma
```

```
telco <- read.csv("telco.csv")
```

```
head(telco)
```

```
##   ID region tenure age  marital address income      ed  
## 1  1 Zone 2      13  44  Married      9      64 College degree
```

```
## 2 2 Zone 3      11 33 Married      7 136 Post-undergraduate degree
## 3 3 Zone 3      68 52 Married     24 116 Did not complete high school
## 4 4 Zone 2      33 33 Unmarried   12 33 High school degree
## 5 5 Zone 2      23 30 Married      9 30 Did not complete high school
## 6 6 Zone 2      41 39 Unmarried   17 78 High school degree
## retire gender voice internet forward custcat churn
## 1 No Male No No Yes Basic service Yes
## 2 No Male Yes No Yes Total service Yes
## 3 No Female No No No Plus service No
## 4 No Female No No No Basic service Yes
## 5 No Male No No Yes Plus service No
## 6 No Female No No No Plus service No
```

No missing values, so we may continue

```
colSums(is.na(telco))
```

```
##      ID region tenure      age marital address income      ed
##      0      0      0      0      0      0      0      0
## retire gender voice internet forward custcat churn
##      0      0      0      0      0      0      0
```

Encoding churn feature for further model application

```
telco <- telco %>%
  mutate(
    churn_event = ifelse(churn == "Yes", 1, 0)
  )
```

Creating the survival object

```
surv_obj <- Surv(time = telco$tenure, event = telco$churn_event)
```

```
surv_obj[1:10]
```

```
## [1] 13 11 68+ 33 23+ 41+ 45 38+ 45+ 68+
```

```
aft_formula <- surv_obj ~ age + marital + address + income + ed + retire +
  gender + voice + internet + forward + custcat + region
```

Listing all the distributions to be tested

```
distributions <- c(
  "extreme",
  "logistic",
  "gaussian",
  "weibull",
  "exponential",
  "rayleigh",
  "loggaussian",
  "lognormal",
  "loglogistic",
  "t"
)

aft_models <- list()

for (dist in distributions) {
  aft_models[[dist]] <- survreg(
```

```

    formula = aft_formula,
    data = telco,
    dist = dist
  )
}

```

The candidate AFT models are compared using log-likelihood, which is shown in the `survreg()` model output. A higher log-likelihood indicates a better fit to the observed survival data.

```

model_comparison <- data.frame(
  Model = names(aft_models),
  LogLik = sapply(aft_models, function(model) as.numeric(logLik(model)))
) %>%
  arrange(desc(LogLik))

```

```
model_comparison
```

```

##           Model      LogLik
## loggaussian loggaussian -1457.012
## lognormal   lognormal  -1457.012
## loglogistic loglogistic -1458.103
## weibull     weibull    -1462.172
## exponential exponential -1467.598
## rayleigh    rayleigh   -1527.438
## gaussian    gaussian   -1547.611
## logistic    logistic   -1554.948
## t           t          -1562.957
## extreme     extreme    -1571.191

```

As a decision maker, I would use the lognormal AFT model. The lognormal and loggaussian models achieved the highest log-likelihood, indicating the strongest statistical fit among the tested distributions. Since these two models produce the same fit in this analysis, I choose the lognormal model because it is the more standard and interpretable name. Beyond statistical fit, the lognormal model is preferable because it allows a flexible churn pattern over tenure, unlike the exponential model, which assumes a constant churn risk. This is important in telecom churn because customers may behave differently at early and later tenure stages. The lognormal AFT model is also useful for CLV because it directly models survival time and can generate customer-level survival predictions needed for lifetime value calculations.

```

# Kaplan-Meier empirical survival curve
km_fit <- survfit(surv_obj ~ 1, data = telco)

km_curve <- data.frame(
  Time = km_fit$time,
  Survival = km_fit$surv,
  Model = "Kaplan-Meier"
)

# Probabilities used for AFT quantile prediction
# p = cumulative churn probability
# survival = 1 - p
p_values <- seq(0.01, 0.99, by = 0.01)

# Function to create average AFT survival curve for one model
get_aft_curve <- function(model, model_name, data, p_values) {

  pred_time <- predict(

```

```

    model,
    newdata = data,
    type = "quantile",
    p = p_values
  )

  # If prediction returns a matrix, average predicted quantile time across customers
  if (is.matrix(pred_time)) {
    avg_time <- colMeans(pred_time, na.rm = TRUE)
  } else {
    avg_time <- as.numeric(pred_time)
  }

  data.frame(
    Time = as.numeric(avg_time),
    Survival = 1 - p_values,
    Model = model_name
  )
}

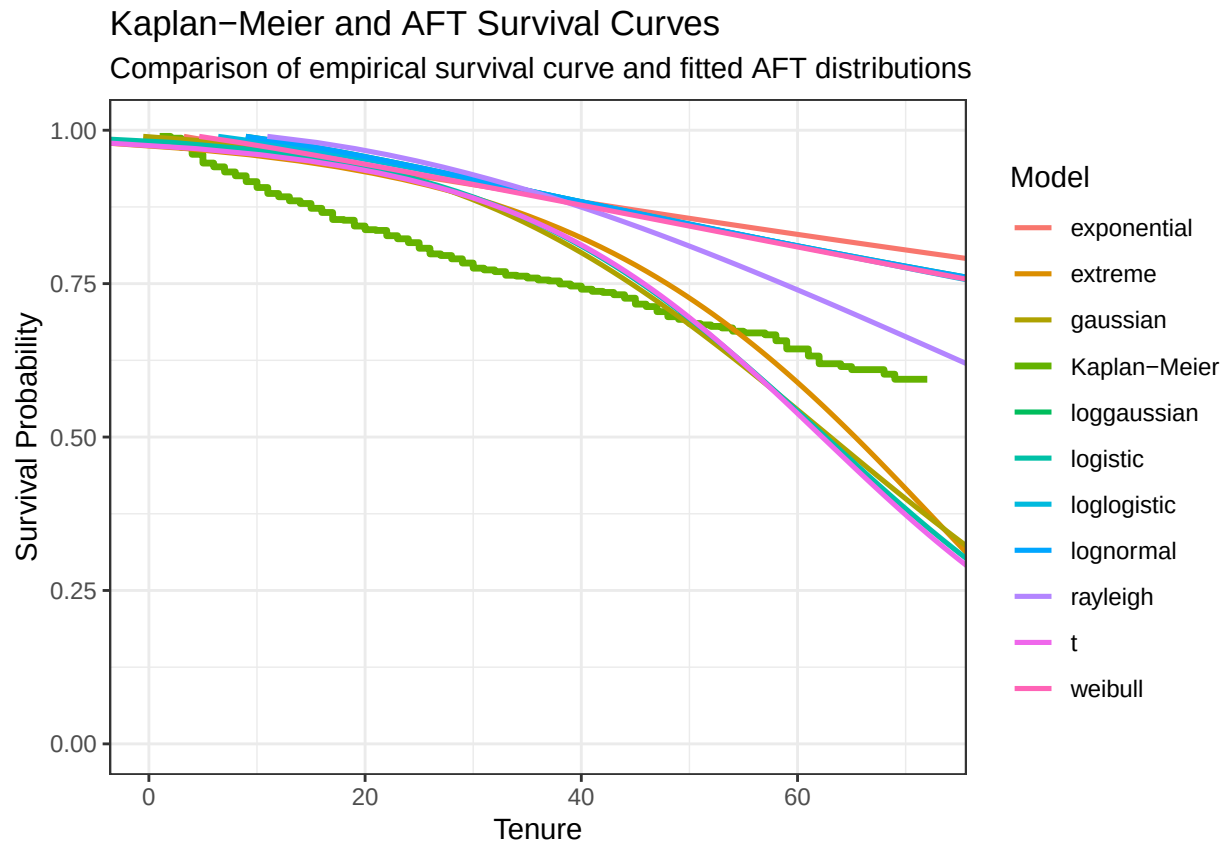
# Generate AFT curves for all fitted distributions
aft_curves <- map_dfr(
  names(aft_models),
  function(model_name) {
    get_aft_curve(
      model = aft_models[[model_name]],
      model_name = model_name,
      data = telco,
      p_values = p_values
    )
  }
)

all_survival_curves <- bind_rows(
  km_curve,
  aft_curves
)

ggplot() +
  geom_step(
    data = km_curve,
    aes(x = Time, y = Survival, color = Model),
    linewidth = 1.2
  ) +
  geom_line(
    data = aft_curves,
    aes(x = Time, y = Survival, color = Model),
    linewidth = 0.9
  ) +
  coord_cartesian(
    xlim = c(0, max(telco$tenure)),
    ylim = c(0, 1)
  ) +

```

```
labs(
  title = "Kaplan-Meier and AFT Survival Curves",
  subtitle = "Comparison of empirical survival curve and fitted AFT distributions",
  x = "Tenure",
  y = "Survival Probability",
  color = "Model"
) +
theme_bw()
```



```
decision_model <- aft_models[["lognormal"]]
```

```
summary(decision_model)
```

```
##
## Call:
## survreg(formula = aft_formula, data = telco, dist = dist)
##
```

	Value	Std. Error	z	p
## (Intercept)	2.36227	0.29263	8.07	6.9e-16
## age	0.03267	0.00725	4.50	6.7e-06
## maritalUnmarried	-0.45515	0.11543	-3.94	8.0e-05
## address	0.04254	0.00890	4.78	1.8e-06
## income	0.00140	0.00092	1.52	0.129
## edDid not complete high school	0.37361	0.20159	1.85	0.064
## edHigh school degree	0.31593	0.16318	1.94	0.053
## edPost-undergraduate degree	-0.03436	0.22317	-0.15	0.878
## edSome college	0.27232	0.16535	1.65	0.100
## retireYes	0.02248	0.44407	0.05	0.960

```
## genderMale          0.05188    0.11429  0.45    0.650
## voiceYes            -0.43379    0.16895 -2.57    0.010
## internetYes         -0.77150    0.14348 -5.38 7.6e-08
## forwardYes          -0.19813    0.18004 -1.10    0.271
## custcatE-service     1.06642    0.17053  6.25 4.0e-10
## custcatPlus service  0.92495    0.21575  4.29 1.8e-05
## custcatTotal service 1.19860    0.25045  4.79 1.7e-06
## regionZone 2        -0.09704    0.14277 -0.68    0.497
## regionZone 3         0.04822    0.14154  0.34    0.733
## Log(scale)          0.27577    0.04600  6.00 2.0e-09
##
## Scale= 1.32
##
## Log Normal distribution
## Loglik(model)= -1457 Loglik(intercept only)= -1602.5
## Chisq= 291.01 on 18 degrees of freedom, p= 3.4e-51
## Number of Newton-Raphson Iterations: 5
## n= 1000
```

```
coef_table <- as.data.frame(summary(decision_model)$table)
```

```
coef_table
```

```
##              Value Std. Error      z
## (Intercept)  2.362269428 0.2926296824  8.07255576
## age          0.032670376 0.0072539391  4.50381168
## maritalUnmarried -0.455150373 0.1154299850 -3.94308613
## address       0.042536388 0.0089035396  4.77746936
## income        0.001396065 0.0009202265  1.51708852
## edDid not complete high school 0.373611261 0.2015875430  1.85334498
## edHigh school degree  0.315927691 0.1631833280  1.93602922
## edPost-undergraduate degree -0.034359588 0.2231726395 -0.15395968
## edSome college  0.272318219 0.1653468311  1.64695154
## retireYes       0.022477360 0.4440694911  0.05061676
## genderMale      0.051876408 0.1142876955  0.45391070
## voiceYes        -0.433794624 0.1689533765 -2.56754042
## internetYes     -0.771503017 0.1434832103 -5.37695676
## forwardYes      -0.198134394 0.1800375528 -1.10051704
## custcatE-service  1.066415595 0.1705321083  6.25345927
## custcatPlus service 0.924945623 0.2157517416  4.28708300
## custcatTotal service 1.198604072 0.2504521577  4.78576061
## regionZone 2    -0.097040714 0.1427699495 -0.67969985
## regionZone 3     0.048223177 0.1415380089  0.34070832
## Log(scale)      0.275774484 0.0459985325  5.99528875
##
##              p
## (Intercept)  6.884172e-16
## age          6.674536e-06
## maritalUnmarried 8.043975e-05
## address       1.775151e-06
## income        1.292443e-01
## edDid not complete high school 6.383293e-02
## edHigh school degree  5.286412e-02
## edPost-undergraduate degree 8.776415e-01
## edSome college  9.956801e-02
## retireYes      9.596309e-01
```

```
## genderMale                6.498931e-01
## voiceYes                   1.024228e-02
## internetYes                7.575536e-08
## forwardYes                 2.711069e-01
## custcatE-service           4.014593e-10
## custcatPlus service        1.810347e-05
## custcatTotal service       1.703410e-06
## regionZone 2               4.966945e-01
## regionZone 3               7.333232e-01
## Log(scale)                  2.031242e-09
```

```
decision_model <- aft_models[["lognormal"]]

# Extract coefficient table
coef_table <- as.data.frame(summary(decision_model)$table) %>%
  rownames_to_column("Coefficient")

# Keep significant coefficients only
significant_coefficients <- coef_table %>%
  filter(
    !Coefficient %in% c("(Intercept)", "Log(scale)"),
    p < 0.05
  )
```

The final reduced model keeps the predictors that were significant in the full lognormal model. Coefficients are interpreted through time ratios: values above 1 mean longer expected survival time, while values below 1 mean shorter expected survival time and higher churn risk.

```
final_formula <- surv_obj ~ age + marital + address + voice + internet + custcat

final_model <- survreg(
  formula = final_formula,
  data = telco,
  dist = "lognormal"
)

summary(final_model)
```

```
##
## Call:
## survreg(formula = final_formula, data = telco, dist = "lognormal")
##
```

	Value	Std. Error	z	p
## (Intercept)	2.53488	0.24261	10.45	< 2e-16
## age	0.03683	0.00640	5.75	8.7e-09
## maritalUnmarried	-0.44732	0.11447	-3.91	9.3e-05
## address	0.04282	0.00885	4.84	1.3e-06
## voiceYes	-0.46350	0.16677	-2.78	0.0054
## internetYes	-0.84054	0.13826	-6.08	1.2e-09
## custcatE-service	1.02582	0.16905	6.07	1.3e-09
## custcatPlus service	0.82250	0.16942	4.85	1.2e-06
## custcatTotal service	1.01326	0.20958	4.83	1.3e-06
## Log(scale)	0.28303	0.04602	6.15	7.7e-10

```
##
## Scale= 1.33
##
## Log Normal distribution
```

```
## Loglik(model)= -1462.1   Loglik(intercept only)= -1602.5
## Chisq= 280.83 on 8 degrees of freedom, p= 4.9e-56
## Number of Newton-Raphson Iterations: 5
## n= 1000

final_coef_table <- as.data.frame(summary(final_model)$table) %>%
  rownames_to_column("Coefficient") %>%
  filter(!Coefficient %in% c("(Intercept)", "Log(scale)")) %>%
  mutate(Time_Ratio = exp(Value))

final_coef_table
```

```
##      Coefficient      Value Std. Error      z      p
## 1      age 0.03682617 0.006399569  5.754477 8.691047e-09
## 2 maritalUnmarried -0.44731642 0.114471837 -3.907655 9.319634e-05
## 3      address 0.04282382 0.008847367  4.840289 1.296502e-06
## 4      voiceYes -0.46349735 0.166770479 -2.779253 5.448407e-03
## 5      internetYes -0.84053742 0.138259182 -6.079433 1.206083e-09
## 6      custcatE-service 1.02582303 0.169046542  6.068288 1.292813e-09
## 7      custcatPlus service 0.82250447 0.169424136  4.854707 1.205651e-06
## 8 custcatTotal service 1.01326065 0.209578546  4.834754 1.333106e-06
##      Time_Ratio
## 1  1.0375127
## 2  0.6393416
## 3  1.0437540
## 4  0.6290797
## 5  0.4314786
## 6  2.7893903
## 7  2.2761934
## 8  2.7545681
```

CLV is calculated using the lecture logic: monthly margin multiplied by the sum of discounted monthly survival probabilities. Here I use a monthly margin of 1300 and an annual discount rate of 10%.

```
# CLV setup from slides
MM <- 1300
r <- 0.10
months <- 1:24

# Predicted log survival time from final lognormal model
lp <- predict(final_model, newdata = telco, type = "lp")
sigma <- final_model$scale

# Survival probabilities
surv_probs <- data.frame(ID = telco$ID)

for (m in months) {
  surv_probs[[paste0("month_", m)]] <- 1 - pnorm((log(m) - lp) / sigma)
}

# Discount survival probabilities
discounted_surv <- surv_probs

for (m in months) {
  discount_factor <- (1 + r / 12) ^ (m - 1)
  discounted_surv[[paste0("month_", m)]] <-
```



```

discounted_surv[[paste0("month_", m)]] / discount_factor
}

telco$CLV <- MM * rowSums(discounted_surv[paste0("month_", months)])

summary(telco$CLV)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      9177  23672   26450   25357   27855   28406

```

```
head(telco$CLV)
```

```
## [1] 26555.99 26699.17 28322.63 24005.53 27209.49 27562.17
```

The next tables compare predicted CLV across customer segments. I use customer category, internet service, and marital status because they are useful for business interpretation and some of them are included in the final model.

```

# CLV by customer category
clv_custcat <- telco %>%
  group_by(custcat) %>%
  summarise(
    n = n(),
    avg_CLV = mean(CLV),
    median_CLV = median(CLV)
  ) %>%
  arrange(desc(avg_CLV))

```

```
clv_custcat
```

```

## # A tibble: 4 x 4
##   custcat      n avg_CLV median_CLV
##   <chr>    <int>   <dbl>    <dbl>
## 1 Plus service    281  27033.    27612.
## 2 E-service      217  26444.    27146.
## 3 Total service  236  24680.    25420.
## 4 Basic service  266  23298.    23824.

```

```

clv_internet <- telco %>%
  group_by(internet) %>%
  summarise(
    n = n(),
    avg_CLV = mean(CLV),
    median_CLV = median(CLV)
  ) %>%
  arrange(desc(avg_CLV))

```

```
clv_internet
```

```

## # A tibble: 2 x 4
##   internet      n avg_CLV median_CLV
##   <chr>    <int>   <dbl>    <dbl>
## 1 No        632  26437.    27301.
## 2 Yes       368  23501.    24365.

```

```

clv_marital <- telco %>%
  group_by(marital) %>%

```

```

summarise(
  n = n(),
  avg_CLV = mean(CLV),
  median_CLV = median(CLV)
) %>%
arrange(desc(avg_CLV))

clv_marital

## # A tibble: 2 x 4
##   marital      n avg_CLV median_CLV
##   <chr>    <int>   <dbl>     <dbl>
## 1 Married    495  26127.    27049.
## 2 Unmarried  505  24601.    25625.

# Survival probability at 12 months
telco$surv_12 <- 1 - pnorm((log(12) - lp) / sigma)

# Churn probability within 12 months
telco$churn_prob_12 <- 1 - telco$surv_12

summary(telco$churn_prob_12)

##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.0000164 0.0159616 0.0645279 0.1105646 0.1710018 0.7563920

I define at-risk customers as the top 25% of subscribers with the highest predicted probability of churn within
the next 12 months.

risk_cutoff <- quantile(telco$churn_prob_12, 0.75)

telco$at_risk_12 <- ifelse(telco$churn_prob_12 >= risk_cutoff, 1, 0)

table(telco$at_risk_12)

##
##    0    1
## 750 250

at_risk_summary <- telco %>%
  filter(at_risk_12 == 1) %>%
  summarise(
    n_at_risk = n(),
    avg_CLV = mean(CLV),
    total_CLV_at_risk = sum(CLV)
  )

at_risk_summary

##   n_at_risk  avg_CLV total_CLV_at_risk
## 1       250 20772.85         5193212

retention_budget_rate <- 0.20

annual_retention_budget <- at_risk_summary$total_CLV_at_risk * retention_budget_rate

annual_retention_budget

```

[1] 1038642

Findings and Recommendations

The lognormal AFT model was selected because it had the highest log-likelihood together with the loggaussian model, while being easier to interpret and more standard for reporting. In the final model, positive coefficients increase expected survival time, while negative coefficients shorten expected survival time and indicate higher churn risk. Age and address have positive effects, meaning older customers and customers who have lived longer at the same address are expected to stay longer. Unmarried customers, customers with voice service, and customers with internet service have shorter expected survival time, meaning they are more likely to churn earlier. Compared with Basic service customers, E-service, Plus service, and Total service customers have much longer expected survival times.

I define valuable customers as customers with high predicted CLV, because they represent higher expected future value after accounting for survival probability and discounting. Based on the segment results, Plus service and E-service customers have the highest average CLV, while Basic service customers have the lowest. Customers without internet service also have higher average CLV than customers with internet service, which is consistent with the model showing that internet service customers have higher churn risk. For the retention budget, I define at-risk subscribers as the top 25% of customers with the highest predicted probability of churn within the next 12 months. This gives 250 at-risk customers with total CLV of 5,193,212. I would allocate 20% of this value as the annual retention budget, giving a budget of 1,038,642. Retention should focus on high-CLV customers who are also at risk, especially through targeted offers, service-quality improvements, and personalized campaigns for internet and voice-service users.